

Development of exosome-encapsulated paclitaxel to overcome MDR in cancer cells

Nanomedicine: Nanotechnology, Biology and Medicine 2016

Abstract record only. No openly available full-text PDF was found for this paper (it appears to be behind a paywall or not deposited in an open repository). This file contains the bibliographic record and the abstract. Access the publisher version via the DOI above, e.g. through your institution.

Journal: Nanomedicine: Nanotechnology, Biology and Medicine

Year: 2016

Citation as given: Nanomedicine. 2016;12(3):655-664

DOI: 10.1016/j.nano.2015.10.012 | **Link:** <https://www.ncbi.nlm.nih.gov/pmc/articles/4809755>

Authors: Kim Myung Soo, Haney Matthew J., Zhao Yuling, Mahajan Vivek, Deygen Irina, Klyachko Natalia L., Inskoe Eli, Piroyan Aleksandr, Sokolsky Marina, Okolie Onyi, Hingtgen Shawn D., Kabanov Alexander V.

Abstract

Exosomes have recently come into focus as “natural nanoparticles” for use as drug delivery vehicles. Our objective was to assess the feasibility of an exosome-based drug delivery platform for a potent chemotherapeutic agent, paclitaxel (PTX), to treat MDR cancer. Herein, we developed and compared different methods of loading exosomes released by macrophages with PTX (exoPTX), and characterized their size, stability, drug release, and in vitro antitumor efficacy. Reformation of the exosomal membrane upon sonication resulted in high loading efficiency and sustained drug release. Importantly, incorporation of PTX into exosomes increased cytotoxicity more than 50 times in drug resistant MDCKMDR1 (Pgp+) cells. Next, our studies demonstrated a nearly complete co-localization of airway-delivered exosomes with cancer cells in a model of murine Lewis Lung Carcinoma pulmonary metastases, and a potent anticancer effect in this mouse model. We conclude that exoPTX holds significant potential for the delivery of various chemotherapeutics to treat drug resistant cancers.